



EASTERN MEDITERRANEAN UNIVERSITY
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DEPARTMENT OF MECHANICAL ENGINEERING

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PROJECT NAME: Hydronic Heating System Design

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SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Abstract	2, 7	1, 2, 3, 4, 5	5		
Introduction and scope	2, 7	1, 2, 3, 4, 5	10		
Building, climate & comfort conditions	2, 7	1, 2, 3, 4, 5	10		
Heat loss calculations	2, 7	1, 2, 3, 4, 5	20		
System calculations	2, 7	1, 2, 3, 4, 5	20		
Pipe sizing & pump selection	2, 7	1, 2, 3, 4, 5	20		
Results & discussion	2, 7	1, 2, 3, 4, 5	5		
References & appendix	2, 7	1, 2, 3, 4, 5	10		
					Marks for report format are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	Acquire a knowledge of thermal environment, comfort and indoor design criteria.	X	X					
2	Understand the climatic design and external design conditions.	X	X					
3	Understand and evaluate thermal transmittance, building heat losses and design heating load.	X	X					X
4	Acquire a knowledge about heating systems and understand their working principles.		X					
5	Learn how to design a hydronic heating system and its components e.g. heat generator selection, pipe and pump sizing etc.	X	X					
6	Understand various parameters associated with cooling load and be able to estimate cooling load of buildings by steady cyclic method.	X						
7	Comprehend the importance of ventilation, distinguish different ventilation systems and be able to carry out simple duct calculations and sizing.	X	X					
8	Learn the methods for air conditioning, comprehend the psychrometric chart and be able to analyze typical air conditioning processes by using psychrometric chart.	X						
	Weight of Student Outcomes	H	H					H

Instructions:

- A. Report presentation:** Write relevant equations and your assumptions. Draw neat, labeled diagrams where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations/standards used. References should be from authentic sources. Include the relevant standards (preferably in the Appendix).
- B. Submission:** Upload your report (as a pdf file) and supplementary files to the "Projects" section in LMS within the designated time. The file uploaded should be named as [MENG443-Project 1-Student ID(s)-Student Name(s)]. No late submissions will be accommodated.

ABSTRACT

The project aims to design a heating system for a building located in Las Vegas, in Nevada USA. To do that, all the building specifications were entered into the design-builder application to determine the heat loss for the whole building and each zone specifically, which was found to be 18.54KW after adding a 10% design margin. Then, the selection of the emitters took place after applying safety factors from the selected catalogue. After that, the hydronic design calculations were conducted including the pipe sizing and flow rates. Lastly, the losses due to the pipe length and fittings of the index circuit were determined to select the appropriate pump. The selected pump is UPM2 25-40180 which is selected to overcome the loss of 0.5960 m of the system by providing the 1.0324 m³/h flow rate needed.

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1. INTRODUCTION AND SCOPE

Background Information:

The term HVAC fundamentally encompasses Heating, Ventilation, and Air Conditioning. Achieving thermal comfort is, in fact, one of the primary objectives when designing HVAC systems. Thermal comfort refers to a mental state in which individuals express subjective approval and contentment with their thermal surroundings. HVAC systems are created to provide indoor thermal conditions that ensure occupants' comfort. There are six major factors that significantly influence how people perceive thermal conditions: air temperature, mean radiant temperature, relative air speed, humidity, metabolic heat production, and clothing insulation[1]. HVAC designs aim to establish an indoor environment that promotes the health, well-being, and overall satisfaction of the people inside.

Heating load calculations within HVAC systems represent a crucial procedure used to size and design heating equipment capable of compensating for the maximum heat loss that a building experiences due to outdoor conditions. Numerous factors affect heating load. For example, the building's geographical location is vital because it defines the characteristics of the surrounding outdoor environment[2]. Additionally, the materials incorporated into the building can have a substantial impact on the required heating load, as various materials and thicknesses possess different thermal properties. Moreover, both infiltration and ventilation contribute considerably to heat loss by allowing cooler outside air to enter the indoor space directly.

Scope of this Project:

This project aims to design a heating system for a specified house under given conditions. The work involves estimating the total heating load resulting from thermal transmission, ventilation, and infiltration, followed by illustrating the calculations and the selection process for the heat emitters, including the heat generator (water boiler). The project then provides the system with the appropriate hydraulic design to ensure effective operation.

This stage includes determining the spacing of the emitters, designing the piping network, performing the necessary pipe-sizing calculations, identifying the circuit with the greatest pressure loss, and choosing a suitable circulation pump to compensate for that loss. Finally, the project concludes with calculating the required expansion vessel size to guarantee the overall stability of the design. Some applications will also be employed for assistance in this project to help illustrate and calculate dimensions and values respectively, softwares like AutoCAD and Design Builder.

2. BUILDING, CLIMATE, AND COMFORT CONDITIONS:

Las Vegas in Nevada, USA was chosen for the location of the building. This location is known for its dry summers and light winters. The building is meant to accommodate less than 10 occupants (7 to 10) and it was oriented at an angle of 29° from the north. It was modeled in the software Design Builder based on the dimensions in the provided AutoCAD file.



Figure 1; layout of the building in DesignBuilder.

The building has two floor (the ground floor and the first floor) and a roof:

- The Ground Floor: This floor consists of an entrance hall, kitchen, cellar, living room and a garage. The figure is shown below.

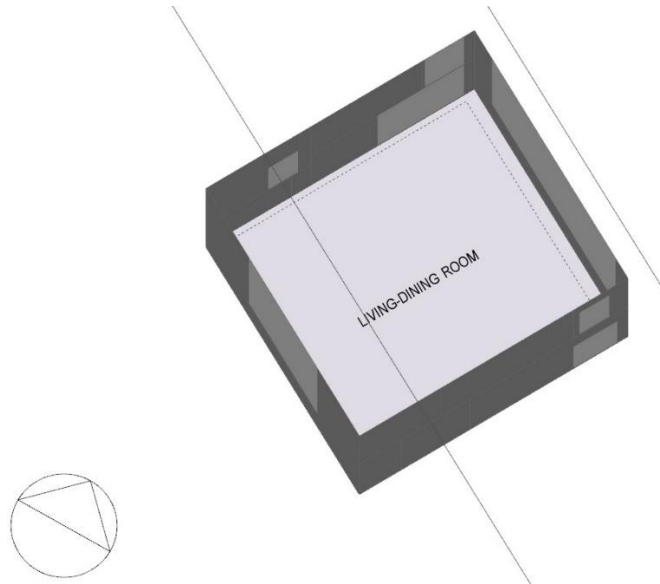


Figure 2; Living room zone in the ground floor.

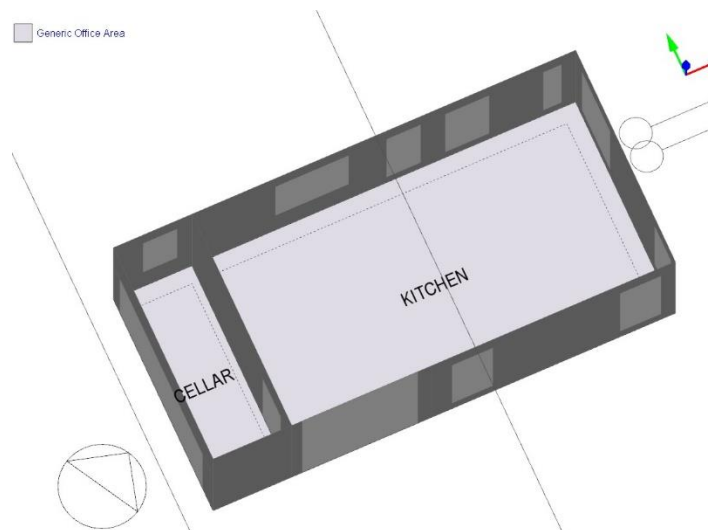


Figure 3; The kitchen and cellar zone.

- The First Floor: Includes a master bedroom with an ensuite bathroom, another master bedroom, 2 extra bedrooms, a hallway and a living area.

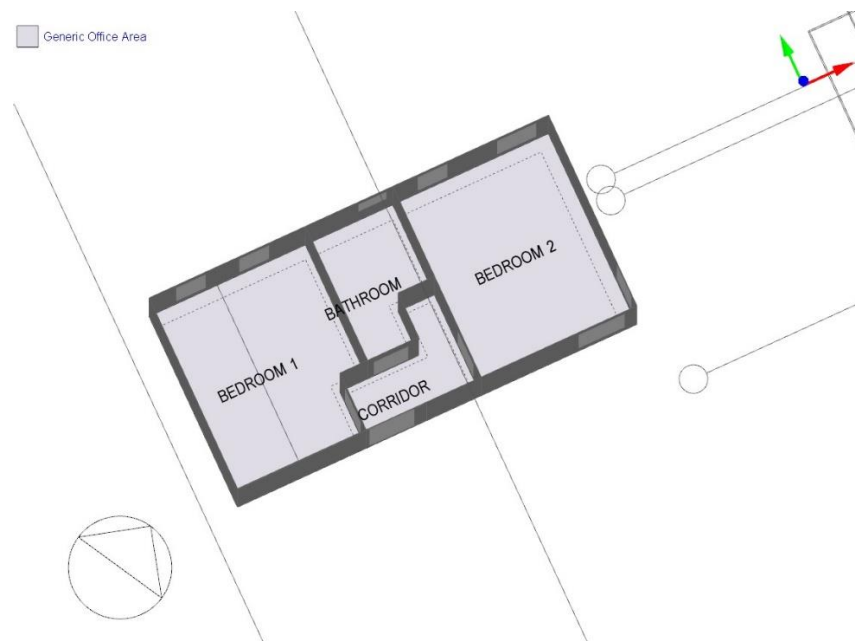


Figure 4; The different zones present in the first floor.

- The Roof: The roof was drawn in approximation with the given dimensions. The inaccuracy of the roof does not have any sizeable impact in the results in calculations.

Internal and External Design Conditions: The outside dry bulb temperature is 0.3°C , the air temperature is 23°C , and all operative temperatures of zones are assumed to be 20°C . The materials are given in the AutoCAD file of the ground floor, first floor, flat roof, pitched roof, walls.

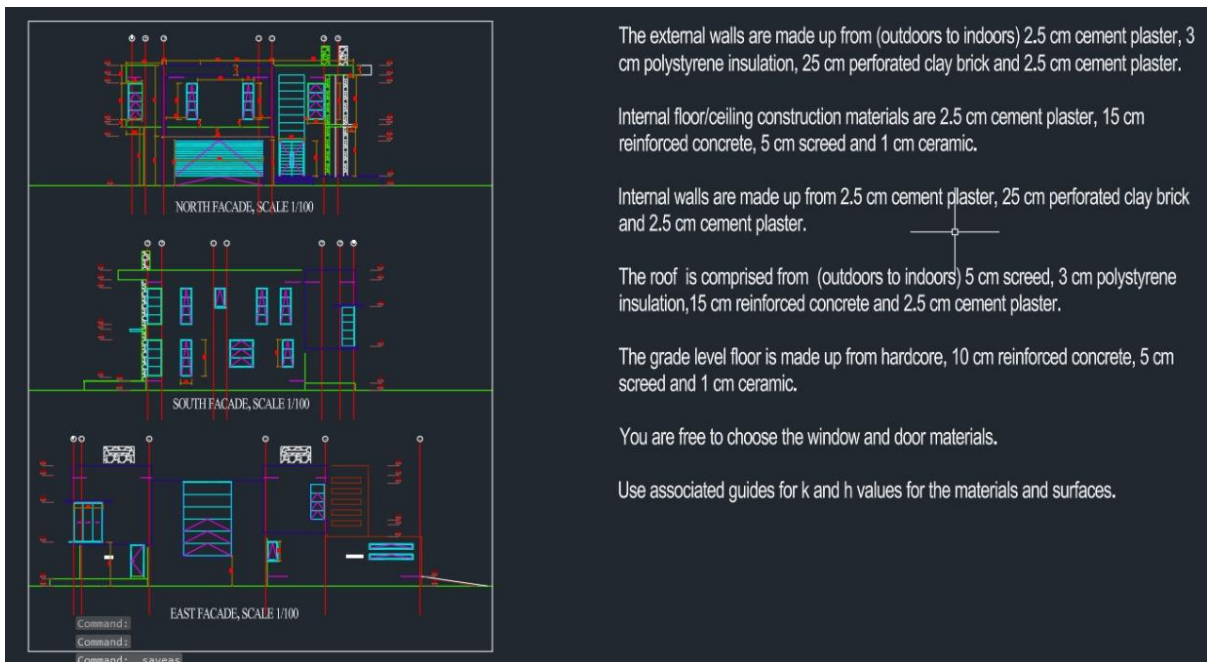


Figure 5; The construction materials in AutoCAD.

N/B: Windows are Double glazing absorptive, no shading (36 windows) and the windows to wall ratio is 40%

Data Report (Not Editable)	
General	
Double glazing, clear, no shading	
Category	Double glazing
Region	General
External Glazing	
Glazing	Dbl Clr 6mm/13mm Air
Frame construction	Painted Wooden window frame
% Glazing area opens	5
Internal Glazing	
Glazing	Sgl Clr 3mm
Frame construction	Painted Wooden window frame
% Glazing area opens	0
Roof Glazing	
Glazing	Sgl Clr 3mm
Frame construction	Painted Wooden window frame
% Glazing area opens	0
External Shading	
Detailed Shading Data	
Window shading	No
Roof window shading	No
Local shading	No

Figure 6; The selected properties of the windows.

Note that the doors are painted oak and 13 in total.

3. HEAT LOSS CALCULATIONS:

The heat-loss calculations follow the procedures outlined in the lecture notes and account for factors such as construction layers and their U-values, occupant activities and schedules, the number of occupants, zone conditioning, lighting power density, and ventilation needs. These inputs were adjusted in Design Builder using reasonable assumptions and relevant standards to achieve accurate model results.

Construction Process:

The number of layers was specified, and materials were either selected directly from Design Builder's library or adapted using data from the CIBSE Guide, as shown in the following figure. Material properties such as density, thermal conductivity, and specific heat capacity were defined accordingly. All materials referenced from the CIBSE Guide either matched those in the AutoCAD file, had equivalent properties under different names (e.g., cast dense reinforced concrete corresponding to 'RC slabs'), or showed only negligible differences (e.g., coarse gravel vs. earth gravel; perforated clay brick assumed to match brickwork inner-leaf properties). Cast concrete was assumed for all concrete components, and the basement ground floor was assumed to be soil. Layer thicknesses were assigned as listed in the following table, consistent with the dimensions provided in the earlier AutoCAD figure.



Figure 7; Edited properties of the external wall.

Table 1: Given Materials and their Thickness

GROUND FLOOR	INTERNAL FLOOR	ROOF	EXTERNAL WALLS
<i>Hardcore Reinforced</i>	Cement Plaster	Screed (5cm)	Cement Plaster
<i>Concrete (10cm)</i>	(2.5cm)		(2.5cm)
<i>Screed (5cm)</i>	Reinforced Concrete	Polystyrene	Polystyrene
	(15cm)	Insulation (3cm)	Insulation (3cm)
<i>Ceramic (1cm)</i>	Screed (5cm)	Reinforced Concrete	Perforated Clay Brick
		(15cm)	(25cm)
	Ceramic (1cm)	Cement Plaster	Cement Plaster
		(2.5cm)	(2.5cm)

Sample Calculations of U value for the external walls:

Assume perforated clay brick to match the brickwork inner-leaf value.

$$R_{se} = \frac{1}{E(hr+hc)}$$

$$\varepsilon = 0.9, K = 1, \varepsilon K = E = 0.9$$

$$h_r = \text{radiative coefficient} = \sigma 4 T_s^3 = 5.67 \times 10^{-8} \times 4 \times (273.15 - 0.5)^3 = 4.59 \text{ W/m}^2\text{K}$$

$$h_c = \text{convective coefficient} = 4 + 4V = 4 + 4 \times 4 \text{ m/s} = 20 \text{ W/m}^2\text{s}^{-1}$$

N/B: External velocity was assumed to be 4 m/s.

$$R_{se} = 0.045 \text{ m}^2\text{K/W}, R_{si} = \frac{1}{\frac{6}{5}E(hr+hc)}$$

$$\varepsilon = 0.9, K = 1, \varepsilon K = E = 0.9$$

$$h_r = \sigma 4 T_s^3 = 5.67 \times 10^{-8} \times 4 \times (273.15 + 23)^3 = 5.89 \text{ W/m}^2\text{K}$$

$$h_c = 2.5 \text{ W/m}^2\text{K}$$

N/B: Indoor air conditions were assumed to be still air conditions (horizontal).

$$R_{si} = 0.11036 \text{ m}^2\text{K/W}$$

$$R = \frac{d}{K}, R_1 = \frac{0.025}{0.72}, R_2 = \frac{0.03}{0.036}, R_3 = \frac{0.25}{0.62}, R_4 = \frac{0.025}{0.72}$$

$$R_{total} = R_{si} + R_{se} + R_1 + R_2 + R_3 + R_4$$

$$R_{total} = 1.459 \text{ m}^2\text{K/W}$$

$$U = 1/R_{total} = 0.685 \text{ W/m}^2\text{K}$$

Table 2; *U values of building sections.*

COMPONENT	U VALUE (W/m²K)	NUMBER OF LAYERS
<i>External wall</i>	0.685	5
<i>Internal wall</i>	1.623	3
<i>Roof</i>	1.026	5
<i>Ground floor</i>	3.076	3
<i>Internal floor</i>	2.382	4
<i>External & internal doors</i>	2.823	1
<i>External windows</i>	3.094	—

ACTIVITY:

All zones on the ground and first floors were assumed to be conditioned, with each zone assigned to its own schedule. For example, the following figure illustrates the schedule for the ground-floor bathroom. As noted earlier, the building houses six occupants, and the estimated conditioning periods for the bedrooms are approximately 9 hours on weekdays and 11 hours on weekends. Refer to the appendix for schedules of the remaining zones.

Table 3; Light power density of each zone.

ZONE	LIGHTING POWER DENSITY W/m²
<i>Bedrooms and kitchen</i>	12 / 13
<i>Bathrooms and lounge</i>	10.5
<i>Entrance hall and hallway</i>	15
<i>Study room</i>	14

For Ventilation & Infiltration: CIBSE guides table was used in calculating the ACH and the equation below:

$$N = \frac{3600 \times q_v}{V}$$

A sample calculation of the kitchen ventilation:

$$q_v = 60\text{L/s} = 0.06\text{m}^3/\text{s}, A = 29.78\text{m}^2, H = 3.28$$

$$N = \frac{3600 \times 0.06}{29.78 \times 3.28} = 2.2 \text{ ACH}$$

Building/room type	Winter operative temp. range for stated activity and clothing levels*			Summer operative temp. range (air conditioned buildings†) for stated activity and clothing levels*			Suggested air supply rate / (L.s ⁻¹ per person) unless stated otherwise	Filtration grade‡	Maintained illuminance¶ / lux	Noise ratings§ (dba)
	Temp. / °C	Activity / met	Clothing / clo	Temp. / °C	Activity / met	Clothing / clo				
Dwellings:										
— bathrooms	20–22	1.2	0.25	23–25	1.2	0.25	15 L.s ⁻¹	G2–G4 (extract)	150	—
— bedrooms	17–19	0.9	2.5	23–25	0.9	1.2	0.4–1 ACH to control moisture	G2–G4	100	25
— hall/stairs/landings	19–24	1.8	0.75	21–25	1.8	0.65	—	—	100	—
— kitchen	17–19	1.6	1.0	21–23	1.6	0.65	60 L.s ⁻¹	G2–G4 (extract)	150–300	40–45
— living rooms	22–23	1.1	1.0	23–25	1.1	0.65	0.4–1 ACH to control moisture	G2–G4	50–300	30
— toilets	19–21	1.4	1.0	21–23	1.4	0.65	>5 ACH	G2–G4	100	—
Educational buildings:										
— lecture halls	19–21	1.4	1.0	21–23	1.4	0.65	10	G4–G5	500	25–35
— seminar rooms	19–21	1.4	1.0	21–23	1.4	0.65	10	G4–G5	300	25–35
— teaching spaces	19–21	1.4	1.0	21–23	1.4	0.65	10	G4–G5	300	25–35
Exhibition halls	19–21	1.4	1.0	21–23	1.4	0.65	10	G3–G4	300	40
Factories:										
— heavy work	11–14	2.5	0.85	—	—	—	—	Depends on use	—	50–65
— light work	16–19	1.8	0.85	—	—	—	—	Depends on use	—	45–55
— sedentary work	19–21	1.4	1.0	21–23	1.4	0.65	—	Depends on use	—	45
Fire/ambulance stations:										
— recreation rooms	20–22	1.3	1.0	22–24	1.3	0.65	10	F5	300	35–40
— watchroom	22–23	1.1	1.0	24–26	1.1	0.65	10	F5	200	35–40

Figure 8; Air supply rate per zone, derived from CIBSE guide.[3]

Table 4; Infiltration rates of each zone.

ZONES	CONSTANT INFILTRATION RATE (ACH)
Bedrooms and halls	0.7
Master bathroom	5.0
Ground & first floor bathroom	3.0
Kitchen	2.2

HEAT LOSS: Here are the calculated heat loss and design capacity for each zone extracted from the Design builder.

Block /	Zone	Comfort Temperature (°C)	Steady-State Heat Loss...	Design Capacity (kW)	Design Capacity [W/m2]	Glazing Gains ...	Wall Gains (kW)	Floor Gains (k...	Roof and Ceil...	Ventilation Gai...	Infiltration Gai...
[-] Building 1 Total Design Heating Capacity = 23.180 (kW)											
[-] Block 1 Total Design Heating Capacity = 2.900 (kW)											
Block 1	GARAGE	19.26	2.32	2.90	83.1886	-0.175	-1.001	0.062	-0.767	0.000	-0.441
[-] Block 2 Total Design Heating Capacity = 2.560 (kW)											
Block 2	GARAGE ENTRANCE	19.16	0.34	0.43	44.6803	0.000	-0.126	-0.016	-0.064	0.000	-0.135
Block 2	WASHROOM	18.92	0.40	0.50	184.5090	-0.032	-0.072	-0.001	-0.022	0.000	-0.274
Block 2	ENTRANCE AND COR...	20.01	1.30	1.63	84.6863	-0.240	-0.564	-0.040	-0.160	0.000	-0.299
[-] Block 3 Total Design Heating Capacity = 5.780 (kW)											
Block 3	LIVING-DINING ROOM	19.12	4.62	5.78	130.2485	-1.704	-0.749	0.089	-0.912	0.000	-1.346
[-] Block 4 Total Design Heating Capacity = 0.230 (kW)											
Block 4	CELLAR	18.20	0.19	0.23	992.7457	0.000	-0.175	0.006	-0.016	0.000	-0.002
[-] Block 5 Total Design Heating Capacity = 2.950 (kW)											
Block 5	CELLAR	18.91	0.27	0.34	63.4223	-0.048	-0.118	-0.003	-0.026	0.000	-0.075
Block 5	KITCHEN	17.44	2.09	2.61	89.8234	-0.561	-0.417	0.145	-0.033	0.000	-1.221
[-] Block 6 Total Design Heating Capacity = 2.820 (kW)											
Block 6	BATHROOM	18.55	0.46	0.58	148.0693	-0.031	-0.101	-0.009	-0.100	0.000	-0.219
Block 6	BEDROOM 1	17.52	0.63	0.79	61.1074	-0.119	-0.113	0.030	-0.267	0.000	-0.161
Block 6	CORRIDOR	18.72	0.23	0.28	97.3386	0.000	-0.098	-0.009	-0.081	0.000	-0.038
Block 6	BEDROOM 2	16.79	0.94	1.17	94.1330	-0.273	-0.286	0.047	-0.270	0.000	-0.155
[-] Block 7 Total Design Heating Capacity = 1.730 (kW)											
Block 7	STUDY ROOM	15.47	1.38	1.73	120.9890	-0.187	-0.250	-0.435	-0.278	0.000	-0.233
[-] Block 8 Total Design Heating Capacity = 4.210 (kW)											
Block 8	MASTER BEDROOM	16.93	1.13	1.41	80.8304	-0.353	-0.346	0.153	-0.366	0.000	-0.219
Block 8	MASTER BEDROOM ...	16.41	1.69	2.11	210.1421	-0.135	-0.197	-0.247	-0.213	0.000	-0.897
Block 8	WARDROBE ROOM	17.62	0.55	0.69	58.3565	-0.059	-0.158	0.062	-0.248	0.000	-0.147

Figure 9; The steady state heat loss and design capacity.



Figure 10; The results Graph of design conditions and heat loss gains.

Air Temperature (°C)	20.32
Radiant Temperature (°C)	16.05
Operative Temperature (°C)	18.18
Outside Dry-Bulb Temperature (°C)	0.30
Glazing (kW)	-3.92
Walls (kW)	-4.77
Ceilings (int) (kW)	-0.32
Floors (int) (kW)	0.32
Ground Floors (kW)	0.24
Partitions (int) (kW)	0.00
Roofs (kW)	-3.50
Floors (ext) (kW)	-0.73
External Infiltration (kW)	-5.86
Zone Sensible Heating (kW)	18.54

Figure 11; Building's main findings from DB.

4. SYSTEM CALCULATIONS

Once the building's overall heating load was determined using Design Builder, the heating load for each individual zone was also identified. This was necessary because each zone will have its own emitter sized to meet its specific heating demand. The selected emitters come from the Caloree Technical catalogue, which features radiative emitters. Radiative emitters were chosen over convective ones to avoid limitations on furniture placement, providing residents with greater flexibility in arranging their living spaces.

The process of selecting the emitters for each zone is as follows:

- The inlet and outlet water temperatures are taken as 90°C and 70°C, respectively.
- In the selected catalogue, the standard design is based on water inlet and outlet temperatures of 90°C/70°C and a zone temperature of 20°C. If any of these conditions differ, correction factors must be applied to the zone heating load to determine the actual emitter size required. For each zone, the load should be multiplied by the appropriate correction factor corresponding to the deviation from the chosen design zone temperature. The recommended zone temperatures from the CIBSE Guide are as follows:

Building/room type	Winter operative temp. range for stated activity and clothing levels*			Summer operative temp. range (air conditioned buildings†) for stated activity and clothing levels*			Suggested air supply rate / (L.s ⁻¹ per person) unless stated otherwise	Filtration grade‡	Maintained illuminance¶ / lux	Noise rating§ (NR)
	Temp. / °C	Activity / met	Clothing / clo	Temp. / °C	Activity / met	Clothing / clo				
Dwellings:										
— bathrooms	20–22	1.2	0.25	23–25	1.2	0.25	15 L.s ⁻¹	G2–G4 (extract) ^[8]	150 ^[4]	—
— bedrooms	17–19	0.9	2.5	23–25	0.9	1.2	0.4–1 ACH to control moisture ^[8]	G2–G4	100 ^[4]	25
— hall/stairs/landings	19–24 ^[4]	1.8	0.75	21–25 ^[1]	1.8	0.65	—	—	100	—
— kitchen	17–19	1.6	1.0	21–23	1.6	0.65	60 L.s ⁻¹	G2–G4 (extract) ^[8]	150–300	40–45
— living rooms	22–23	1.1	1.0	23–25	1.1	0.65	0.4–1 ACH to control moisture ^[8]	G2–G4	50–300	30
— toilets	19–21	1.4	1.0	21–23	1.4	0.65	>5 ACH	G2–G4	100 ^[4]	—

Figure 12; CIBSE guide zones properties recommendations.

Table 5; Selected design temp for each zone.

ZONE	DESIGN TEMPERATURE (°C)
Entrance hall	22
Garage	22
Kitchen	19
Entrance & Corridor	22
Garage entrance	20
Washroom	20
Livingroom & Dining room	22
Cellar	20
Bathroom	20
Bedroom 1	19
Bedroom 2	19
Corridor	20
Study room	18

<i>Master bedroom</i>	19
<i>Master bedroom's WC & bathroom</i>	19
<i>Wardrobe room</i>	19

Afterward, the correction factor from the selected catalogue was determined for every zone and applied.

Sample selection of the correction factor for the garage:

The design-builder load for the entrance is 2.32 kW and the design temperature is 22°. The inlet-outlet temperatures are 90°C and 70°C respectively. From the table, the actual required load is then

Correction factor $f @ 22^{\circ} = 0.937$

$$2.32 \times 0.937 = 2.17\text{kW}$$

Table 6; The approach of selecting the correction factor from the catalogue.

f factor parameters could be available to calculate different working temperatures else than 90/70-20 °C

WORKING TEMPERATURE tg°C tç°C		tr °C						
		10	15	18	20	22	24	26
90	85	1.412	1.290	1.218	1.171	1.124	1.073	1.033
	80	1.348	1.228	1.157	1.111	1.065	1.019	0.974
	75	1.284	1.165	1.095	1.049	1.004	0.959	0.914
	70	1.219	1.102	1.032	1.000	0.937	0.897	0.853
	65	1.154	1.037	0.968	0.943	0.879	0.835	0.791
	60	1.087	0.971	0.903	0.893	0.814	0.771	0.728

**FOR DIFFERENT
TEMPERATURES (f) CORRECTION
FACTOR TABLE**

$$f = \left[\frac{t_g - t_c}{60 \ln \left[\frac{t_g - t_o}{t_c - t_o} \right]} \right]^n \quad Q_n = Q_h \times f$$

f : Correction factor
 Qn : New heat output
 n : A-verace; 1,35
 Qh : Heat output at (90/70-20°C)
 tg= Inletwater temperature.°C
 tc= Outletwater temperature.°C
 to= Room temperature.°C

Figure 13; Formula used in obtaining correction factor.

In the same way, the load of each zone was corrected as follows:

Table 7; Corrected load to the catalogue design.

ZONE	DESIGNBUILDER LOAD (KW)	CORRECTION FACTOR (f)	REQUIRED LOAD (KW)
Washroom	0.40	1	0.40
Livingroom	4.62	0.937	4.33
Cellar	0.46	1	0.46
Kitchen	2.09	1.016	2.12
Bathroom	0.46	1	0.46
Bedroom 1	0.60	1.016	0.61
Bedroom 2	0.94	1.016	0.96

<i>Study room</i>	1.38	1.032	1.42
<i>Master bedroom</i>	1.13	1.016	1.15
<i>Master WC & bathroom</i>	1.69	1.016	1.72
<i>Wardrobe room</i>	0.55	1.016	0.56

- After that, the emitters were selected using the corrected load. The zones were divided into three categories, small, medium, and large zones. For small zones that include the cellar bathrooms, type 11 was selected for its small size. For medium zones, which include the study room, ground and first halls, bedroom 1, and bedroom 2, type 22 was selected for the moderate size. The rest are large zones where type 33 emitters were selected for larger systems e.g. the living room.
- For each type, there were tables where the size of the emitter could be selected according to the inlet-outlet temperatures and the required load. The selection of the emitters was performed.

Sample emitter selection for living room:

The required load for the living room would be 4.33 KW. The selected emitter must be able to satisfy the needed load. Note that the selected emitter size is not to exceed at least 1200 mm (1.2 m). Using all of that and the selected inlet outlet temperatures, the size of the emitter was selected as type 33-DKEK 1000mm x 900mm which gave a load of 4,366W.

Table 8; Emitter selection for the living room.

20 °C Room temperature (*) Water inlet-outlet temperature

TYPE 33-DKEK	HEIGHT (*)	300			400			500			600			750			900		
		HEAT OUTPUT			HEAT OUTPUT			HEAT OUTPUT			HEAT OUTPUT			HEAT OUTPUT			HEAT OUTPUT		
LENGTHS	°C	Watt	Kcal/h	BTU	Watt	Kcal/h	BTU	Watt	Kcal/h	BTU	Watt	Kcal/h	BTU	Watt	Kcal/h	BTU	Watt	Kcal/h	BTU
300	55/45	220	189	750	277	238	946	330	284	1126	379	326	1293	447	385	1527	516	444	1760
	75/65	429	369	1465	545	468	1859	653	561	2227	754	648	2573	890	765	3036	1025	881	3498
	90/70	545	469	1860	693	596	2366	833	716	2842	964	829	3290	1137	978	3880	1310	1126	4469
400	55/45	293	252	1000	369	318	1261	440	378	1501	505	434	1724	597	513	2035	688	591	2347
	75/65	572	492	1953	726	625	2479	870	748	2970	1006	865	3431	1186	1020	4047	1367	1175	4664
	90/70	727	625	2481	925	795	3155	1110	955	3789	1286	1105	4387	1516	1304	5173	1746	1502	5959
500	55/45	366	315	1249	462	397	1576	550	473	1877	631	543	2155	746	641	2544	860	739	2934
	75/65	716	615	2441	908	781	3098	1088	936	3712	1257	1081	4289	1483	1275	5059	1709	1469	5830
	90/70	909	781	3101	1156	994	3944	1388	1193	4736	1607	1382	5484	1895	1629	6466	2183	1877	7449
600	55/45	439	378	1499	554	477	1891	660	567	2252	758	652	2586	895	769	3053	1032	887	3520
	75/65	859	738	2930	1090	937	3718	1306	1123	4455	1508	1297	5147	1779	1530	6071	2050	1763	6996
	90/70	1090	938	3721	1387	1193	4732	1666	1432	5683	1929	1658	6580	2274	1955	7759	2620	2252	8939
700	55/45	513	441	1749	647	556	2206	770	662	2627	884	760	3017	1044	898	3562	1204	1035	4107
	75/65	1002	861	3418	1271	1093	4338	1523	1310	5197	1760	1513	6005	2076	1785	7083	2392	2057	8162
	90/70	1272	1094	4341	1618	1391	5521	1943	1671	6630	2250	1935	7677	2653	2281	9053	3056	2628	10428
800	55/45	586	504	1999	739	635	2521	880	757	3003	1010	869	3448	1193	1026	4071	1376	1183	4694
	75/65	1145	984	3906	1453	1249	4957	1741	1497	5940	2011	1729	6862	2372	2040	8095	2734	2350	9327
	90/70	1454	1250	4961	1849	1590	6310	2221	1909	7577	2571	2211	8774	3032	2607	10346	3493	3003	11918
900	55/45	659	567	2249	831	715	2837	990	851	3378	1137	977	3879	1342	1154	4580	1548	1331	5281
	75/65	1288	1107	4394	1634	1405	5577	1958	1684	6682	2263	1945	7720	2669	2295	9107	3075	2644	10493
	90/70	1636	1406	5581	2080	1789	7099	2498	2148	8525	2893	2487	9870	3411	2933	11639	3929	3379	13408
1000	55/45	732	630	2499	924	794	3152	1100	946	3753	1263	1086	4310	1491	1282	5088	1720	1479	5867
	75/65	1431	1230	4883	1816	1561	6196	2176	1871	7425	2514	2162	8578	2965	2550	10119	3417	2938	11659
	90/70	1817	1563	6201	2312	1988	7887	2776	2387	9472	3214	2764	10967	3790	3259	12932	4366	3754	14898
1100	55/45	806	693	2749	1016	874	3467	1210	1040	4129	1389	1195	4740	1640	1410	5597	1891	1626	6454
	75/65	1574	1353	5371	1998	1718	6816	2394	2058	8167	2765	2378	9436	3262	2805	11131	3759	3232	12825
	90/70	1999	1719	6822	2543	2186	8676	3053	2626	10419	3536	3040	12064	4169	3585	14226	4803	4130	16387
1200	55/45	879	756	2999	1108	953	3782	1320	1135	4504	1516	1303	5171	1790	1539	6106	2063	1774	7041
	75/65	1717	1477	5859	2179	1874	7436	2611	2245	8910	3017	2594	10294	3559	3060	12142	4100	3526	13991
	90/70	2181	1875	7442	2774	2385	9465	3331	2864	11366	3857	3316	13161	4548	3911	15519	5239	4505	17877

The same procedure was followed and the emitters for the other zones were selected.

Table 9; The selected emitters for each building zone.

ZONE	TYPE USED	EMITTER SIZE (mm)	LOAD (KW)
Washroom	11-PK	500 x 400	0.410
Livingroom	33-DKEK	1000 x 900	4.366
Cellar	11-PK	500 x 500	0.496
Kitchen	33-DKEK	800 x 500	2.221
Bathroom	11-PK	500 x 500	0.496
Bedroom 1	22-PKKP	500 x 300	0.637
Bedroom 2	22-PKKP	600 x 400	0.973
Study room	22-PKKP	900 x 400	1.460

<i>Master bedroom</i>	33-DKEK	700 x 300	1.272
<i>Master WC & bathroom</i>	22-PKKP	900 x 500	1.749
<i>Wardrobe room</i>	11-PK	600 x 500	0.596

Boiler Selection:

The selected boiler should satisfy the total system load after selecting the emitters. The load of the boiler must be more than the sum of all emitters loads. Since the DesignBuilder load was 18.54KW, the boiler selected should be in the range of 19 – 20KW. The selected boiler the job was 20KW in size [4].



Figure 14; Selected boiler (Vaillant ecoTEC Plus 620 20kW System Boiler 0010036011)

Table 10; Specifications of the selected boiler.

 Specifications	
Technical	
Weight	34 Kg
MPN	0010036011
Product Code	PN0010036011
Boiler/Product Type	System Boiler
Sub Type	Gas System
Guarantee (Years)	5
Width (mm)	440
Height/Length (mm)	720
Depth (mm)	352
Heat Output	20 kW
Fuel Type	Natural Gas
Mounting	Wall
ErP Rating	ErP A

Mass flowrate of the system:

The mass flow rate calculations are needed to perform the hydraulic design and pipe sizing. The following equation was used to determine the flow rate for each emitter.

$$\dot{m} = \dot{Q} / C_p \Delta T$$

$\dot{m}$ = The mass flow rate of the emitter (kg/s).

$\dot{Q}$ = Emitter heating load (kW).

C_p = Heat capacity of water (kJ/kg °C).

ΔT = Temperature difference of the inlet and return water (°C).

The heat capacity of water at 80°C (mean temperature) is 4.198 kJ/(kg °C).

Sample calculation for the flowrate of the washroom emitter:

The emitter load is 0.410 kW, and the flowrate is

$$\dot{m} = \frac{0.410}{4.198 \times 20} = 0.00488 \frac{\text{Kg}}{\text{s}}$$

The flow rates for the other emitters were calculated and listed in the following table accordingly.

Table 11; Calculated mass flow rates for the emitters.

<i>ZONE</i>	<i>EMITTER LOAD (KW)</i>	<i>FLOW RATES (Kg/S)</i>
<i>Washroom</i>	0.410	0.00488
<i>Livingroom</i>	4.366	0.0520
<i>Cellar</i>	0.496	0.00591
<i>Kitchen</i>	2.221	0.0265
<i>Bathroom</i>	0.496	0.00591
<i>Bedroom 1</i>	0.637	0.00759
<i>Bedroom 2</i>	0.973	0.0116
<i>Study room</i>	1.460	0.0174
<i>Master bedroom</i>	1.272	0.0152

<i>Master WC & bathroom</i>	1.749	0.0208
<i>Wardrobe room</i>	0.596	0.00709

The total calculated flowrate of the emitters is $0.17488 \frac{Kg}{s}$

$$0.17488 \frac{Kg}{s} = 0.629568 \frac{m^3}{h}$$

The placing of the emitters:

In determining the placement of the emitters for each zone, several restrictions were taken into account. First, the dimensions of the emitters relative to the wall were considered, ensuring that the wall width would never be smaller than the width of the emitter itself. Second, it was required that each emitter be positioned in the coldest area of the zone so that heat could effectively reach all points within the space. Lastly, the piping system for each emitter was evaluated, and the chosen location had to allow for the shortest possible pipe length in order to minimize significant pressure drops along the piping. The recommended emitter locations are presented in the figures included in the Appendix section.

Pipe system:

Before designing the pipe system, the boiler's location needed to be established. Due to factors such as safety, the appearance of the house, and efficient use of space, the option of placing the boiler at the front of the house was dismissed. Positioning the boiler on either side of the house was feasible, but it would lead to a longer piping layout, which in turn would increase the pressure loss in the index circuit and require a higher-capacity pump. For these reasons, the boiler was ultimately placed behind the house. The design process followed several considerations. One was that pipes could not pass through doors, windows, or structural columns for safety purposes. Another was that pipe lengths had to be kept

as short as possible to reduce pressure drops throughout the system. Based on these points, the piping layout was developed. The figures that follow illustrate the final design, with dimensions provided in centimeters

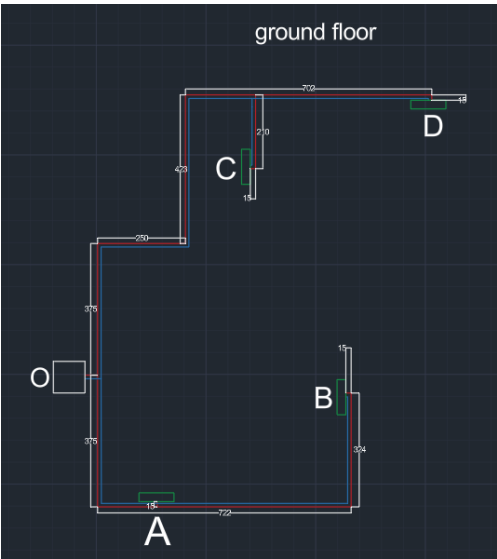


Figure 15; Proposed piping system layout for the ground floor.

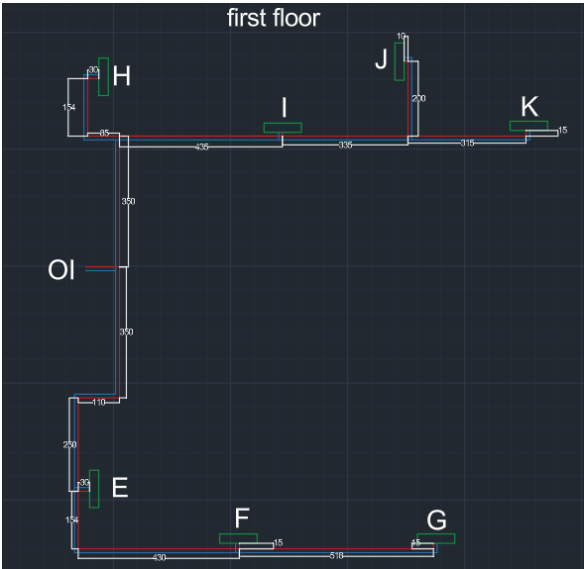


Figure 16; Proposed piping system layout for the first floor.

5. PIPE SIZING AND PUMP SELECTION

pipe sizing and pressure loss:

After designing the piping network, the sizes of the pipes were determined. CIBSE guide approach was followed. The water velocity in a pipe is not to exceed 1 m/s. The pressure loss per meter should not exceed 300 Pa/m as stated in the CIBSE guide. However, the maximum pressure loss selected for the system is 400 Pa/m to avoid very low velocities which can cause system cavitation and are difficult to control precisely [5]. First, copper pipes were selected as the material of our system pipe network . CIBSE Pipe Sizing V2.2 Excel generated Pipe sizing tables for inlet and outlet temperatures[6]. Flowrates in every pipe determine the pipe sizing considering the restriction stated before.

Simple calculation for supply pipe O to C :

The flow rate in that section is equal to the sum of the flow rate in emitters O and C.

$$\dot{m}_{o-c} = 0.00591 + 0.0265 = 0.03241 \frac{kg}{s}$$

Then, the flowrate was entered into the excel sheet and the temperature was set to the inlet temperature, the velocity was determined to be 0.37 m/s, while the pressure loss is 176 Pa/m. And the overall loss as 2196.48 Pa.

Table 12 : Pipe sizing for the supply pipes.

Pipe section	Length (m)	Flowrate (kg/s)	nominal size (mm)	Velocity (m/s)	pressure loss per meter (Pa/m)	pressure loss (Pa)
O-A	5.6	0.02568	12	0.29	118	660.8
A-B	8.91	0.04712	15	0.34	113	1006.83
O-C	12.48	0.03241	12	0.37	176	2196.48
C	2.25	0.0591	15	0.42	168	378
C-D	5.02	0.0265	12	0.3	124	622.48
O-O1	3.17	0.1263	22	0.41	97	307.49
01-E	7.4	0.0208	12	0.24	82	606.8
E-F	5.99	0.0137	10	0.23	105	628.95
F-G	5.33	0.0152	10	0.26	126	671.58
O1-H	3.5	0.05739	15	0.41	160	560
H	2.69	0.0174	10	0.3	159	427.71
H-I	4.45	0.0399	15	0.28	85	378.25
I-J	3.35	0.0324	12	0.37	176	589.6
J	2.1	0.0208	12	0.24	82	172.2
J-K	3.3	0.0116	10	0.2	79	260.7
first pipe	1	0.2787	28	0.53	113	113

Table 13 : Pipe sizing for the Return pipes.

Pipe section	Length (m)	Flowrate (kg/s)	nominal size (mm)	Velocity (m/s)	pressure loss per meter (Pa/m)	pressure loss (Pa)
A-O	5.6	0.02568	12	0.29	138	772.8
B-A	8.91	0.04712	15	0.34	133	1185.03
C-O	12.48	0.03241	12	0.37	196	2446.08
C	2.25	0.0591	15	0.42	168	378
D-C	5.02	0.0265	12	0.3	124	622.48
O1-O	3.17	0.1263	22	0.41	117	370.89
O1-E	7.4	0.0208	12	0.24	118	873.2
F-E	5.99	0.0137	10	0.23	105	628.95
G-F	5.33	0.0152	10	0.26	156	831.48
H-O1	3.5	0.05739	15	0.41	180	630
H	2.69	0.0174	10	0.3	179	481.51
I-H	4.45	0.0399	15	0.28	102	453.9
J-I	3.35	0.0324	12	0.37	186	623.1
J	2.1	0.0208	12	0.24	142	298.2
K-J	3.3	0.0116	10	0.2	119	392.7
A-O	5.6	0.02568	12	0.29	138	772.8

Water pump calculations:

Afterwards, the index circuit, which is the supply return circuit with the most pressure loss from the boiler, was determined. For our case, the circuit with the most length and fittings is assumed to be the index circuit. The supply return from the boiler to emitter c is selected. The pipe pressure loss for that section is highlighted in the previous tables while the fitting calculation will take place in the following section. From the tables, the total pipe pressure loss in the index circuit is 4642.56 Pa.

Fittings loss:

Assumptions taken are that all the bend are smooth; there is a gate valve in the inlet of every supply branch. Lastly, all tees are assumed to be screwed.

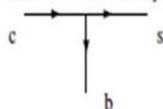
Sample calculations for diverging T (supply):

The diameter for that section is already found to be 12 mm and the velocity is 0.37 m/s. According to the flow shape, the loss was selected from the table of the CIBSE guide. The used table is listed.

Table 14: : CIBSE guide fitting loss table (Diverging Tees).

Diverging flows: Values must be used with the velocity pressure of the combined flow.

(d) $\zeta_{c,b}$ To the branch



d_b/d_c	q_b/q_c									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.3	2.80	4.50	6.00	7.88	9.40	11.1	13.0	15.8	20.0	24.7
0.44	1.41	2.00	2.50	3.20	3.97	4.95	6.50	8.45	10.8	13.3
0.52	1.37	1.81	2.30	2.83	3.40	4.07	4.80	6.00	7.18	8.90
0.59	1.10	1.54	1.90	2.35	2.73	3.22	3.80	4.32	5.28	6.53
0.66	1.22	1.45	1.67	1.89	2.11	2.38	2.58	3.04	3.84	4.75
0.74	1.09	1.20	1.40	1.59	1.65	1.77	1.94	2.20	2.68	3.30
1.00	0.90	1.00	1.13	1.20	1.40	1.50	1.60	1.80	2.06	2.80

$$\frac{d_b}{d_c} = \frac{12}{15} = 0.8$$

$$\frac{q_b}{q_c} = \frac{0.0324}{0.0591} = 0.548$$

Then, interpolation was used to find the value of ξ as 1.5. the density ρ of water at $90^\circ = 965.3 \text{ kg/m}^3$.

$$\Delta p = \frac{1}{2} \rho c^2 \xi = 99.11218 \text{ Pa.}$$

Table 15: Fitting losses in the supply and return line of the index circuit.

Item	Quantity	D(mm)	db/dc	qb/qc	ξ	c (m/s)	$\frac{1}{2} \rho c^2$	$\Delta p = \frac{1}{2} \xi \rho c^2$ (Pa)
Gate valve	1	12			0.03	0.37	66.07479	1.982244
bends 1 (O-C)	8	12			0.55	0.37	528.5983	290.7291
bends 2 (C)	4	15			0.55	0.42	340.5578	187.3068
converging tee	1	12	0.8	0.548	1.1	0.42	85.13946	93.65341
diverging tee	1	12	0.8	0.548	1.5	0.37	66.07479	99.11218
								SUM = 672.784

Knowing that the total pressure loss from the pipe length is 4642.56 Pa, and after adding the fittings loss, the total loss for the index circuit is found to be 5315.34 Pa. Note That 10% is added to account for the losses inside the emitters and boiler. The new total loss is now 5846.87 Pa.

Pump selection :

For the pump selection, GRUNDFOS website was used to determine the possible pumps that can be used according to the specifications for every system.

The UPM2 25-40180 Pump is selected since it is found that it would overcome the maximum pressure loss of 0.5960 m providing the required flow rate of 1.0324 m³/h. The selected pump pdf specification report is attached with the submission file, while some of the specifications are listed. The process of entering values into the website is found in the appendix part. Using affinity laws, the system curve was generated and listed below.



Figure 17: UPM2 25-40 180 pump[7]

Performance

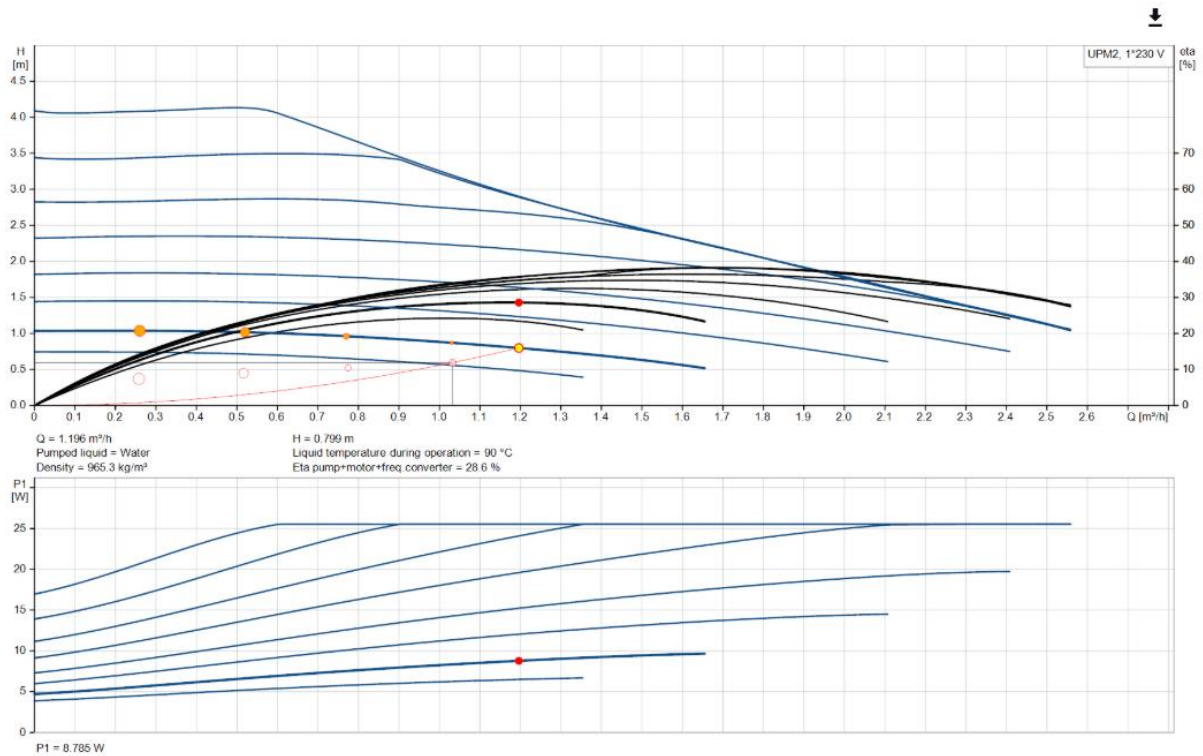


Figure 18 : Pump curve.

Sizing results

Type	UPM2
Quantity	1
Flow	1.195 m³/h (+16%)
Head	0.799 m (+34%)
Min.inlet pressure	0.72 bar (90 °C, against atmosphere)
Power P1	0.00895 kW
Eta pump+motor	28.6 % =Eta pump * Eta motor
Eta total	28.6 % =Eta relative to the duty point
Energy consumption	45 kWh/year
CO2 emission	25 kg/Year
Price	On request

Load profile ①

	1	2	3	4
Flow (%)	25	50	75	100
Flow (m³/h)	0.26	0.52	0.77	1.03
Head (%)	174	171	161	146
Head (m)	1.038	1.018	0.961	0.87
P1 (kW)	0.00571	0.00674	0.00769	0.0085
Eta total (%)	12.6	20.9	25.9	28.3
Time (h/a)	3010	2394	1026	410
Energy consumption (kWh/Year)	17	16	8	3
Quantity	1	1	1	1

Figure 19 : Some pump specifications.

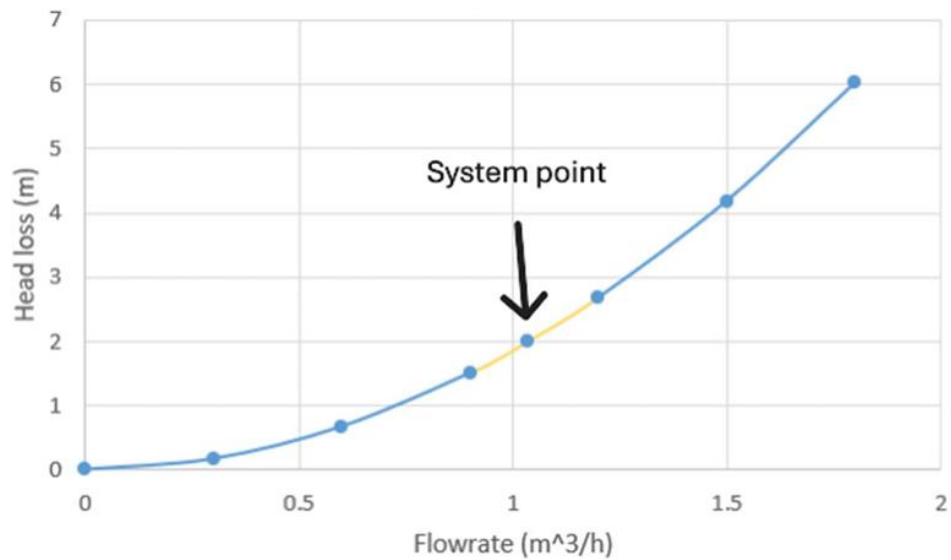


Figure 20 : system curve

Expansion Vessel :

The size of the expansion vessel needed to balance the system is calculated as follows:

- The water volume inside the system is approximated by finding the volume inside every emitter from the catalogue (check appendix) and the water inside the pipes. The total volume of water is 93.43 L. Findings are listed in the following tables.

Table 16; Individual Pipe volume data.

Nominal size	size of the pipe (mm)	liters size of the pipe (mm) volume per meter	Length (m)	Volume(l)
12	8.8	0.0607904	77.86	4.73314054
15	11.3	0.10023665	34.72	3.48021649
22	18	0.25434	3.17	0.8062578
23	20	0.314	1	0.314

Table 17: Water volume inside emitters.

ZONE	TYPE USED	EMITTER SIZE	VOLUME (L)
Washroom	11-PK	500 x 400	2.9
Livingroom	33-DKEK	1000 x 900	9.13
Cellar	11-PK	500 x 500	4.2
Kitchen	33-DKEK	800 x 500	12.43
Bathroom	11-PK	500 x 500	2.25
Bedroom 1	22-PKKP	500 x 300	4.68
Bedroom 2	22-PKKP	600 x 400	4.5
Study room	22-PKKP	900 x 400	4.8
Master bedroom	33-DKEK	700 x 300	8.82
Master WC & bathroom	22-PKKP	900 x 500	2.61
Wardrobe room	11-PK	600 x 500	3.8

- Initial water temperature is taken to be 4°C.
- System height is approximately 3 meters.
- Temperature of the plant is assumed to be 20°C

$$P = \rho gh = 1000 \times 9.81 \times 3 = 29.43 \text{ kPa (gauge)} = 129.43 \text{ (abs)}$$

The expansion of the water from 4°C to 90°C is 3.63%, the used table is listed in the appendix.

$$\Delta V = 93.43 \times \frac{3.63}{100} = 3.39 \text{ L}$$

As the maximum allowed pressure is 350 kPa, at the pump inlet,

$$P = 350 - 19.498 = 330.502 \text{ kPa (gauge)} = 430.502 \text{ (abs)}$$

Then by using ideal gas equation

$$\frac{430.502(V - 3.39)}{293} = \frac{129V}{277}$$

$$V = 4.86 \text{ L}$$

So, to balance the system, an expansion vessel of size 4.86 liters is needed. However, the selected boiler already has an internal expansion vessel of 8 liters, which would be sufficient for the system.

6. RESULTS AND DISCUSSION

After modeling the building in the design-builder, the heat load requirement was calculated. Afterward, the emitter sizes for every zone were determined by applying the correction factors associated with the selected catalogue to account for the design zone temperature. After that, the boiler size was determined by summing up all the emitter loads, then an appropriate catalogue was selected. Then, the pipe sizing calculations took place after finding the flow rate in every section and after placing the boiler and the emitters. Following that, the index circuit was determined, and the pump specifications were found according to that. Lastly, to ensure the system balance the calculation for the size of the expansion vessel was stated. The main findings are listed below.

Table 18; Tabulated summary of the report.

Item	Conclusion
boiler	The required load is 18.54KW. The selected boiler load is 20 kW.
Pressure loss of the index circuit	Found to be 0.59601 Pa = after finding the pipe loss and the fittings losses and add a safety margin.
Total flowrate	1.0324 m ³ /h
Expansion vessel	expansion vessel of size 4.86 liters is needed. However, the selected boiler already has an internal expansion vessel of 8 liters

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Available: <https://product-selection.grundfos.com/au/products/up-oem/upm2-oem/upm2-59C23103?pumpsystemid=2836581924&tab=variant-curves>

APPENDIX

Appendix A: Wall Material Properties

Thermal conductivity, density and specific heat capacity: inorganic, porous materials					
Material	Condition/test (where known)	Source	Thermal conductivity /W·m ⁻¹ ·K ⁻¹	Density /kg·m ⁻³	Specific heat /J·kg ⁻¹ ·K ⁻¹
Asbestos-related materials:					
— asbestos cement		S	1.02	1750	840
— asbestos cement building board		D	0.6	1920	840
— asbestos cement decking		C	0.36	1500	1050
— asbestos cement sheet	Conditioned	C	0.36	700	1050
— asbestos fibre	At 50°C	I	0.06	640	840
— asbestos mill board	At 50°C	I	0.25	1400	840
Brick					
		D	0.72	1920	840
		D	1.31	2080	921
— aerated		S	0.30	1000	840
— brickwork, inner leaf		C	0.62	1700	800
— brickwork, outer leaf		C	0.84	1700	800
— burned		S	0.75	1300	840
		S	0.85	1500	840
		S	1.00	1700	840
— mud	At 50°C	I	0.75	1730	880
— paviour		E	0.96	2000	840
— reinforced	At 50°C	I	1.10	1920	840
— tile	At 50°C	I	0.8	1890	880

Figure 21; Brickwork wall property.

Thermal conductivity, density and specific heat capacity: inorganic, porous materials					
Material	Condition/test (where known)	Source	Thermal conductivity /W·m ⁻¹ ·K ⁻¹	Density /kg·m ⁻³	Specific heat /J·kg ⁻¹ ·K ⁻¹
Cement/plaster/mortar:					
— cement		D	0.72	1860	840
— cement blocks, cellular		T	0.33	520	2040
— cement fibreboard, magnesium oxysulphide binder		A	0.082	350	1300
— cement mortar		S	0.72	1650	920
	Dry	T	0.93	1900	840
	Moist	T	1.5	1900	840
— cement/lime plaster		S	0.8	1600	840
— cement panels, wood fibres	Dry	T	0.08	350	1890
	Moist	T	0.12	350	3040
		T	0.12	400	1470
	Dry	T	0.35	1650	840
— cement plaster		S	0.72	1760	840
		S	1.50	1900	840
— cement plaster, sand aggregate		A	0.72	1860	840
— cement screed		E	1.40	2100	850

Figure 22; Cement plaster ground floor property.

Miscellaneous materials:					
— aggregate	Undried	D	1.8	2240	840
— aggregate (sand, gravel or stone)	Oven dried	A	1.3	2240	920
— building board, tile and lay-in panel		A	0.058	290	590
— calcium silicate brick		S	1.50	2000	840
— granolithic		E	0.87	2085	840
— mud phuska	At 50°C	I	0.52	1620	880
— tile bedding		E	1.40	2100	650
— tile hanging		C	0.84	1900	800
Roofing materials:					
— built-up roofing		D	0.16	1120	1470
— roof tile		C	0.84	1900	800
— tile, terracotta		T	0.81	1700	840

Figure 23; Roof tile properties for the flat roof.

Thermal conductivity, density and specific heat capacity: inorganic, porous materials — <i>continued</i>					
Material	Condition/test (where known)	Source	Thermal conductivity / W·m ⁻¹ ·K ⁻¹	Density / kg·m ⁻³	Specific heat / J·kg ⁻¹ ·K ⁻¹
Concrete, cast:					
— aerated		E	0.16	500	840
		S	0.29	850	840
		S	0.42	1200	840
— aerated, cellular		S	0.15	400	840
		S	0.23	700	840
		S	0.70	1000	840
		S	1.20	1300	840
— aerated, cement/lime based		S	0.21	580	840
— cellular		T	0.16	480	840
	At 50°C	I	0.19	700	1050
— cellular bonded		T	0.30	520	2040
— dense		S	1.70	2200	840
— compacted,		S	2.20	2400	840
— dense, reinforced		S	1.90	2300	840
— compacted		S	2.30	2500	840
— expanded clay filling		S	0.26	780	840
		S	0.60	1400	840

Figure 24; Reinforced concrete property.

Appendix B: Hydraulic system layout indicating pipe sizes.

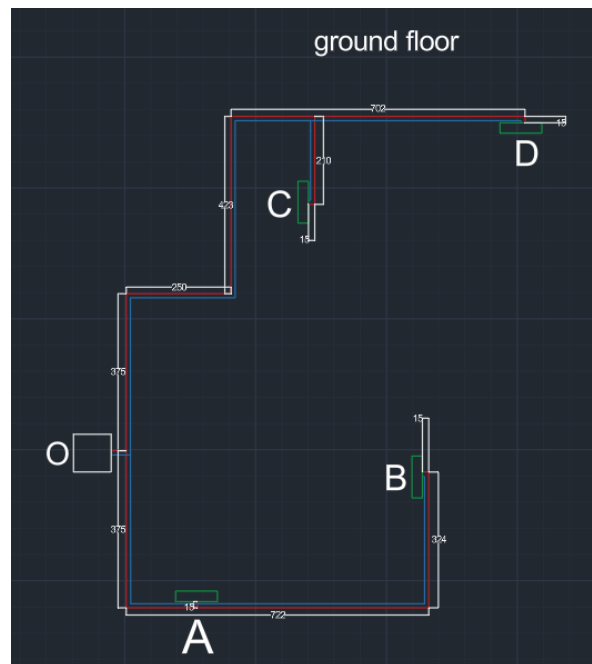


Figure 25: Pipes sizes of the ground floor.

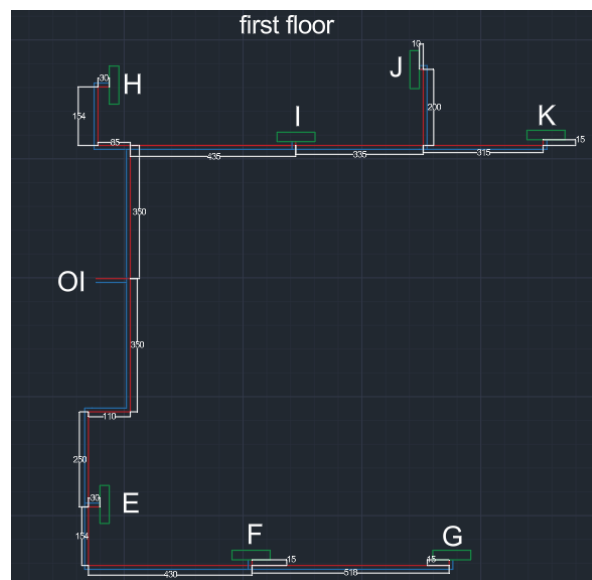


Figure 26: Pipes sizes of the first floor.

Appendix C: Pump selection procedure:

The screenshot shows the Grundfos 'Size Product' web application. The top navigation bar includes links for 'About us', 'Solutions', 'Sustainability', 'Media', 'Careers', and 'Contact'. The main header area is titled 'Size Product' and contains four dropdown menus: 'Size by' (Application), 'Select application area' (Domestic buildings), 'Select application' (Domestic heating), and 'Select type of installation' (Main circulator pump - Domestic Building). Below these is a link 'Open Advanced Selection >'. A row of buttons includes 'Choose action', 'Load parameters', 'Save parameters', 'Reset to defaults', and 'Start sizing'. The 'Select parameters' section includes input fields for 'Flow (Q) *' (1.032434657 m³/h), 'Head (H) *' (0.596012069 m), 'BMS connectivity' (checkbox), 'Evaluation criterion' (Preference index), and 'Prefer fast delivery' (checkbox). The 'Optional sizing parameters' section is titled 'YOUR REQUIREMENTS (Pumped liquid, Min. liquid temperature, Max. liquid temperature, Liquid temperature d...)' and contains input fields for 'Pumped liquid' (Heating water), 'Min. liquid temperature *' (20 °C), 'Max. liquid temperature *' (90 °C), 'Liquid temperature during operation *' (90 °C), 'Min. inlet pressure *' (1.5 bar), 'Allowed flow undersize' (0 %), and 'Allowed flow oversize' (10 %).

Size Product

Size by: Application | Select application area: Domestic buildings | Select application: Domestic heating | Select type of installation: Main circulator pump - Domestic Building | [Open Advanced Selection >](#)

Choose action: [Load parameters](#) | [Save parameters](#) | [Reset to defaults](#) | [Start sizing](#)

Select parameters

Flow (Q) *: 1.032434657 m³/h | Head (H) *: 0.596012069 m | BMS connectivity: ☐ | Evaluation criterion: Preference index | Prefer fast delivery: ☐

Optional sizing parameters

YOUR REQUIREMENTS (Pumped liquid, Min. liquid temperature, Max. liquid temperature, Liquid temperature d...)

Pumped liquid: Heating water | Min. liquid temperature *: 20 °C | Max. liquid temperature *: 90 °C | Liquid temperature during operation *: 90 °C | Min. inlet pressure *: 1.5 bar | Allowed flow undersize: 0 % | Allowed flow oversize: 10 %

Figure 27: flow information